

F18A VDP REGISTERS (VR)

The 9918A and F18A have a 14-bit address counter and TI numbers the MS-bit as zero. Address bits are A0..A13, data k

	Same as 9918A Different from 9918A			F18A Only F18A Ignores - reserved 0				
Register	0 MSbit	1	2	3	4	5	6	7 LSbit
VR0 >00	0	0	0	IE1	0	M4	M3	ExtVid
VR1 >01	4K/16K	BLANK	IE0	M1	M2	0	SIZE	MAG
VR2 >02	0	0	0	0	A0	A1	A2	A3
VR3 >03	A0	A1	A2	A3	A4	A5	A6	A7
VR4 >04	0	0	0	0	0	A0	A1	A2
VR5 >05	0	A0	A1	A2	A3	A4	A5	A6
VR6 >06	0	0	0	0	0	A0	A1	A2
VR7 >07	FG0	FG1	FG2	FG3	BG0	BG1	BG2	BG3
VR10 >0A	0	0	0	0	A0	A1	A2	A3
VR11 >0B	A0	A1	A2	A3	A4	A5	A6	A7
VR15 >0F	0	CNT_RESET	CNT_SNAP	CNT_START	S0	S1	S2	S3
VR19 >13	IL0	IL1	IL2	IL3	IL4	IL5	IL6	IL7
VR24 >18	0	0	SPS0	SPS1	T2PS0	T2PS1	T1PS0	T1PS1

VR25 >19	HTO0	HTO1	HTO2	HTO3	HTO4		HPO0	HPO1	HPO2
VR26 >1A	VT00	VT01	VT02	VT03	VT04		VPO0	VPO1	VPO2
VR27 >1B	HTO0	HTO1	HTO2	HTO3	HTO4		HPO0	HPO1	HPO2
VR28 >1C	VT00	VT01	VT02	VT03	VT04		VPO0	VPO1	VPO2
VR29 >1D	SPGS0	SPGS1	HPSIZE2	VPSIZE2	TPGS0	TPGS1	HPSIZE1	VPSIZE1	
VR30 >1E	0	0	0	SPRTMAX0	SPRTMAX1	SPRTMAX2	SPRTMAX3	SPRTMAX4	
VR31 >1F	BML_EN	BML_PRI	BML_TRANS	BML_FAT	BMLPS0	BMLPS1	BMLPS2	BMLPS3	
VR32 >20	BMLBA0	BMLBA1	BMLBA2	BMLBA3	BMLBA4	BMLBA5	BMLBA6	BMLBA7	
VR33 >21	BMLX0	BMLX1	BMLX2	BMLX3	BMLX4	BMLX5	BMLX6	BMLX7	
VR34 >22	BMLY0	BMLY1	BMLY2	BMLY3	BMLY4	BMLY5	BMLY6	BMLY7	
VR35 >23	BMLW0	BMLW1	BMLW2	BMLW3	BMLW4	BMLW5	BMLW6	BMLW7	
VR36 >24	BMLH0	BMLH1	BMLH2	BMLH3	BMLH4	BMLH5	BMLH6	BMLH7	
VR47 >2F	DPM	AUTO INC	PR0	PR1	PR2	PR3	PR4	PR5	
VR48 >30	SIGN-BIT	INC0	INC1	INC2	INC3	INC4	INC5	INC6	
VR49 >31	TILE2_EN	ROW30	ECMT0	ECMT1	Y_REAL	0	ECMS0	ECMS1	
	Tile Map 2 enable		ECM (T)ile		real sprite Y-coord		ECM (S)prite		
	24 / 30 tile rows								
VR50 >32	VR_RESET	GPU_HTRIG	GPU_VTRIG	TL1_OFF	RPTMAX	SCANLINES	POS_ATTR	TILE2_PRI	
	GPU vsync trig		GPU hsync trig		Report sprite max vs 5th sprite		Position vs name attributes		
	Reset VDP registers		Disable all tile 1 layers		Show virtual scan lines				
VR51 >33	0	0	STOPSPRT0	STOPSPRT1	STOPSPRT2	STOPSPRT3	STOPSPRT4	STOPSPRT5	
VR54 >36	GPU0	GPU1	GPU2	GPU3	GPU4	GPU5	GPU6	GPU7	
VR55 >37	GPU8	GPU9	GPU10	GPU11	GPU12	GPU13	GPU14	GPU15	
VR56 >38	0	0	0	0	0	0	0	GPU OP	

VR57 >39	0	0	0	1	1	1	0	0
F18A STATUS REGISTERS (SR)								
SR0	F	5S	C	SP0	SP1	SP2	SP3	SP4
SR1	ID0	ID1	ID2	0	0	0	BLANKING	HF
SR2	GPU ST	GDATA0	GDATA1	GDATA2	GDATA3	GDATA4	GDATA5	GDATA6
SR3	SCANLINE0	SCANLINE1	SCANLINE2	SCANLINE3	SCANLINE4	SCANLINE5	SCANLINE6	SCANLINE7
SR4	NANO2	NANO3	NANO4	NANO5	NANO6	NANO7	NANO8	NANO9
SR5	0	0	0	0	0	0	NANO0	NANO1
SR6	MICRO2	MICRO3	MICRO4	MICRO5	MICRO6	MICRO7	MICRO8	MICRO9
SR7	0	0	0	0	0	0	MICRO0	MICRO1
SR8	MILLI2	MILLI3	MILLI4	MILLI5	MILLI6	MILLI7	MILLI8	MILLI9
SR9	0	0	0	0	0	0	MILLI0	MILLI1
SR10	SECOND8	SECOND9	SECOND10	SECOND11	SECOND12	SECOND13	SECOND14	SECOND15
SR11	SECOND0	SECOND1	SECOND2	SECOND3	SECOND4	SECOND5	SECOND6	SECOND7
SR12	Unused							
SR13	Unused							
SR14	VMAJOR0	VMAJOR1	VMAJOR2	VMAJOR3	VMINOR0	VMINOR1	VMINOR2	VMINOR3
SR15	READ0	READ1	READ2	READ3	READ4	READ5	READ6	READ7
Sprite attribute byte when F18A is locked (compatible with original VDP)								
	ECLK	0	0	0	COL0	COL1	COL2	COL3
Sprite attribute byte when F18A is unlocked, all modes and ECM settings								
	ECLK	FLIP X	FLIP Y	SIZE	PS0	PS1	PS2	PS3
Tile attribute byte in Enhanced Color Modes								
Color Table becomes Tile Attribute Table Base Address and goes from 32 to 256 or 2K (same as name table) bytes								
	PRI	FLIP X	FLIP Y	TRANS	PS0	PS1	PS2	PS3
When T40/T80 Mode, ECM0, and Position-Based Attributes is Enabled								
	FG0	FG1	FG2	FG3	BG0	BG1	BG2	BG3

F18A GPU NEW AND MODIFIED INSTRUCTION SUMMARY

Instruction	Opcode	Addressing		New name
CALL	>0C80	0000 1100 10Ts SSSS	new	CALL
RET	>0C00	0000 1100 0000 0000	new	RET
PUSH	>0D00	0000 1101 00Ts SSSS	new	PUSH
POP	>0F00	0000 1111 00Td DDDD	new	POP
SLC	>0E00	0000 1110 00Ts SSSS	new	SLC
CKON	>03A0		modified	SPI_EN
CKOF	>03C0		modified	SPI_DS
IDLE	>0340		modified	IDLE
LDCR	>3000		modified	SPI_OUT
STCR	>3400		modified	SPI_IN
RTWP	>0380		modified	RTWP
XOP	>2C00		modified	PIX

UNIMPLEMENTED INSTRUCTIONS

SBO
 SBZ
 TB
 BLWP
 STWP
 LWPI
 LIM
 RSET
 LREX

bits are D0..D7.

Mode = M4 M3 M2 M1

GM1 = 0 0 0 0

GM2 = 0 1 0 0

MCM = 0 0 1 0

T40 = 0 0 0 1 (40 col)

T80 = 1 0 0 1 (80 col)

IE1 = Horizontal interrupt enable, see VR19

Name Table Base Address, 1K boundaries

768-bytes per table for 24 rows

960-bytes per table for 30 rows

Color Table Base Address, 64-byte boundaries

32-bytes long for original color mode (default)

For the enhanced color modes this becomes Tile Attribute Table Base Address

The Tile Attribute Table is 256 bytes long

768 to 4K when in position-attribute mode depending on the size of the horz / vert pages.

Pattern Generator Base Address, 2K boundaries

2K for 256 tiles

4K for 2-bit color mode, 6K for 3-bit color mode

Sprite Attribute Table Base Address, 128-byte boundaries

128-bytes (32 sprites, 4 bytes each)

Sprite Pattern Generator Base Address, 2K boundaries

2K for 256 patterns

4K for 2-bit color mode, 6K for 3-bit color mode

Text color / background color

Name Table 2 Base Address, 1K boundaries

768-bytes per table for 24 rows

960-bytes per table for 30 rows

Color Table 2 Base Address, 64-byte boundaries

Works the same as VR3 in Enhanced Color Modes / Position-Attribute Mode

Status Reg (0..15) to read, see table below

counter: reset: 1 = reset, start/stop: 1 = start, 0 = stop, snap = 1 snapshot the counter without stopping

Horizontal Interrupt scan line, 0 to disable, VR0 IE1-bit must = 1

Extra palette-select bits for original color modes

Sprite, Tile-2, and Tile-1 layers

Tile-2 horizontal scroll offset

Tile-2 vertical scroll offset

Tile-1 horizontal scroll offset

Tile-1 vertical scroll offset

HPSIZE = horizontal page size, 0 = 1 page, 1 = 2 pages

VPSIZE = vertical page size, 0 = 1 page, 1 = 2 pages

SPGS = sprite pattern generator offset size, 11=256, 10=512, 01=1K, 00=2K

TPGS = tile pattern generator offset size, 11=256, 10=512, 01=1K, 00=2K

SPRTMAX0..4 = Max sprites per scan line, set to 0 to reset sprite max to jumper setting

BML: Enable, Priority over sprites, Transparent, Fat pixels, Palette select

Bitmap Layer Base Address, 64-byte boundaries

Bitmap x

Bitmap y

Bitmap width

Bitmap height

DPM (data-port mode), 1 to enable

AUTO INC = auto inc palette index after each write

Palette register #

SIGNED two's-complement increment amount for VRAM address, defaults to 1

ECM = enhanced color mode, (T)ile and (S)prite

0=normal/original color

1=1-bit color mode, tile attributes

2=2-bit color mode, tile attributes

3=3-bit color mode, tile attributes

0 = TL2 always on top, 1 = TL2 vs sprite priority is considered

0 = per name attributes in ECMs, 1 = per position attributes

0 = no virtual scan lines, 1 = enable virtual scan lines

0 = report 5th sprite, 1 = report max sprite (VR30)

0 = normal, 1 = disable GM1, GM2, MCM, T40, T80

0 = disable, 1 = trigger GPU on VSYNC

0 = disable, 1 = trigger GPU on HSYNC

0 = ignored, 1 = reset all VDP registers to power-on defaults

Stop Sprite to limit the total number of sprites to process, defaults to 32

GPU address MSB

GPU address LSB, writing resets and loads, then triggers the GPU

GPU Operation performed when register is written:

0 = reset and load PC, 1 = trigger GPU to begin execution

Enable registers over 7, write 000111xx to VR57 twice

ID = >00 for 9938, >40 (010xxxxx) for 9958

ID = >E0 (111xxxxx) for F18A

HF = 1 when scan line == R19, if R19 != 0

BLANKING = 1 during horz or vert blanking periods

GPU Status: 1 = running, 0 = idle

GPU Data: 7-bits of user programmable data from the GPU

Current scan line, 0 when not in active area

Nanosecond counter (0..999) LSB

Nanosecond counter (0..999) MSB

Microsecond counter (0..999) LSB

Microsecond counter (0..999) MSB

Millisecond counter (0..999) LSB

Millisecond counter (0..999) MSB

Second counter (0..65535) LSB

Second counter (0..65535) MSB

Major/Minor version, 00011001 = v1.9

VDP read register value (see VR15). Updated any time a VRAM address is set

Size: 0 = global VR1 setting, 1 = 16x16, Palette select per sprite

s long

PRI = priority over sprites

TRANS = when 1, colors with 0, 00 or 000 will be transparent instead of a palette index

T40/T80 fg / bg color



Call subroutine, push return address on stack
Return from subroutine, pop return address from stack
Push a 16-bit word onto the stack
Pop a 16-bit word off of the stack
Shift Left Circular
Sets the chip enable line to the SPI Flash ROM low (enables the ROM)
Sets the chip enable line to the SPI Flash ROM high (disables the ROM)
Forces the GPU state machine to the idle state, restart with VR56 trigger
Writes a byte (always a byte operation) to the SPI Flash ROM
Reads a byte (always a byte operation) from the SPI Flash ROM
Does not use R13, only performs R14->PC, R15->status flags
New dedicated pixel plotting and addressing instruction

